Conics: Problem Solving

Questions 1–11

Question 1. The curve C and the line L have equations $y^2 = 12x$ and $y = 6 - 4x$ respectively.

a) Sketch C and L on the same axes.

b) Find the points of intersection between the curve C and the line L

c) Find the exact length of the line segment between the two points of intersection.

Question 2. Two curves C_1 and C_2 have equations $y^2 = 40x$ and $\frac{x^2}{4} + \frac{y^2}{9} = 1$ respectively.

a) Sketch C_1 and C_2 on the same axes.

b) Find the points of intersection between the curve C_1 and the curve C_2

Question 3. Two curves C_1 and C_2 have equations $y^2 = -2x$ and $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively.

a) Sketch C_1 and C_2 on the same axes.

b) Find the points of intersection between the curve C_1 and the curve C_2

Question 4. A curve C and the line L have equations $xy = 16$ and $y = -kx + 8\sqrt{2}$ respectively.

a) Sketch C

b) Find the x-values of the points of intersection between the curve C and the line L in terms of k

c) Find the value of k such that $y = -kx + 8\sqrt{2}$ is a tangent to the curve.

Prerequisite knowledge for Q5-Q6: Volume of Revolution

Question 5. A hyperbola is given by equation

$$\frac{x^2}{a^2} - \frac{y^2}{3a^2} = 1$$

where a is a positive constant.

a) Write down the equations of the asymptotes of the curve in terms of a

b) Sketch the curve.

The finite region bounded by the curve, the positive x-axis, the line $x = a$ and the line $x = 2a$ is rotated 360° about the x-axis.

c) Find the volume of the solid generated in terms of a, giving your answer in exact form.

Question 6. An ellipse is given by equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

a) Given that $a > b$, sketch the curve.

The finite region bounded by the curve, the positive x-axis and the positive y-axis is rotated 360° about the y-axis.

b) Find the volume of the solid generated in terms, giving your answer in exact form.

Question 7. A parabola P has equation

$$y^2 = 12(x - k)$$

where $k > 0$
The line $y = mx$ is a tangent to P
Without using Calculus, prove that $m = \pm\sqrt{\frac{3}{k}}$

Question 8. An ellipse E has equation

$$\frac{x^2}{4} + \frac{y^2}{16} = 1$$

The line with equation $y = mx + 5$ is a tangent to E. Without using Calculus, find the possible values of m

Question 9. An ellipse E_1 is given by the equation

$$(x-5)^2 + \frac{y^2}{9} = 1$$

A second ellipse E_2 is given by the equation

$$\frac{x^2}{9} + (y-5)^2 = 1$$

a) Sketch E_1 and E_2 on the same axes.

A tangent line of E_1 is of the form $x + y = k$

b) Find the possible values of k

c) Show that this tangent line is also a tangent to E_2

Question 10. Find the equation of the locus of points such that the distance from the point $(0, -4)$ is always equal to the distance from the line $y = 2$, giving your answer in the form

$$x^2 = p(y+q)$$

where p and q are constants to be found.

Question 11. Find the equation of the locus of points such that the distance from the point $(9, 0)$ is always three times the distance from the line $x = 1$